

Trigonometry

Week 1C

Overview

Trigonometry

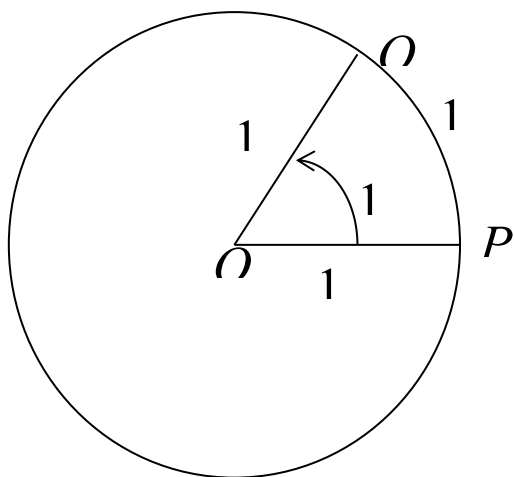
Key Ideas - Radian Measure

- *Exact trigonometric ratio of Angles of any Magnitude (Radians)*
- *Solving Trigonometric Equations in Radians*

Key Idea – Radian Measure

So far, we have been using degrees as our unit of measurement for angles. Measurements in degrees goes back thousands of years and it may have arisen from the idea that there were roughly 360 days in a year. Degrees are a convenient unit for everyday practical measurement but when dealing with trigonometric functions another unit, called the *radian*, is more appropriate. **Radian** measure is related to the **radius** of a circle and has more significance in mathematics.

One radian is defined to be the angle subtended at the centre of a unit circle by a unit arc length on the circumference.



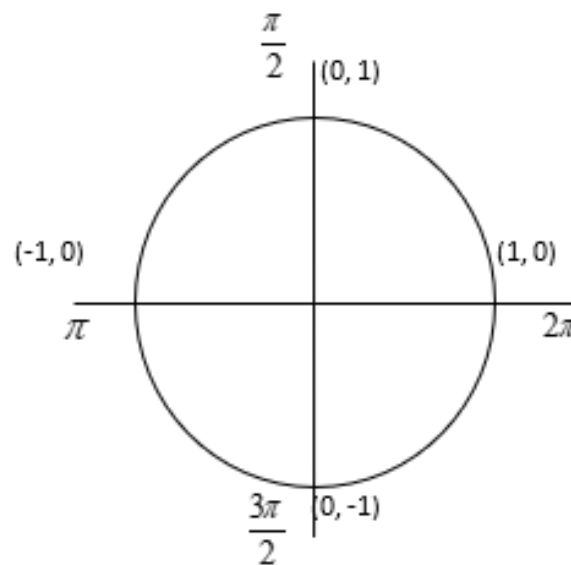
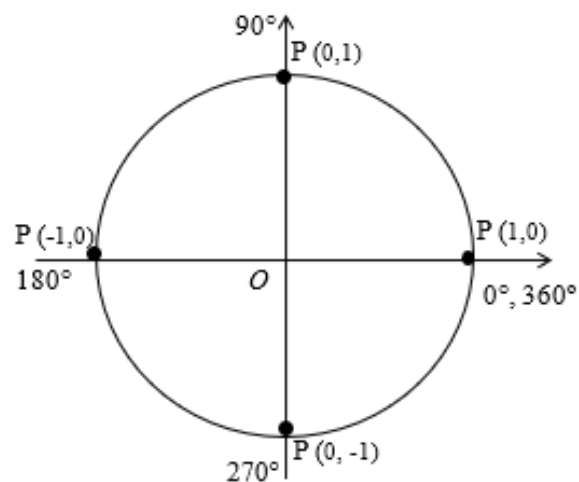
If the rotation is **anticlockwise** the angle is taken as **positive**; if the rotation is **clockwise** the angle is taken as **negative**.

The **circular measure** of angles involves the measurement of angles in **radians**.

Conversion Between Units

Given the circumference of a circle $C = 2\pi r$, if $r = 1$ unit then $C = 2\pi$. Therefore, the whole circumference of a circle is 2π and one revolution is equal to 2π radians.

Hence 2π radians $= 360^\circ$
and π radians $= 180^\circ$



$\pi \text{ radians} = 180^\circ$

It is important to remember this conversion!

$1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ$

$1 \text{ degree} = \frac{\pi}{180} \text{ radians}$

$$1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ \approx 57^\circ 18'$$

$$1 \text{ degree} = \frac{\pi}{180} \text{ radians} \approx 0.01745^c$$

Therefore:

- To convert from radians to degrees: Divide by π , multiple by $180^\circ \rightarrow$ multiply by $\frac{180}{\pi}$
- To convert from degrees to radians: Divide by 180° , multiply by $\pi \rightarrow$ multiply by $\frac{\pi}{180}$

Note: When no units are indicated, the radian is understood to be the unit of angle measurement being used. Sometimes the superscript c is used for radians (circular measure).

Example:

Express 90° in radians.

Remember $180^\circ = \pi$ radian.

$$\therefore 1^\circ = \frac{\pi}{180}$$

$$90^\circ = \frac{90 \times \pi}{180} \text{ radians}$$

$$= \frac{\pi}{2} \text{ radians}$$

← in exact form

$$= 1.57^\circ$$

← to 2 decimal places

Exercise:

(a) Express the following in radians in exact form:

$$30^\circ = \frac{30^\circ}{180^\circ} \pi = \frac{\pi}{6}$$

$$135^\circ = \frac{135^\circ}{180^\circ} \pi = \frac{3\pi}{4}$$

(b) Express 156° in radians to 2 decimal places:

$$156^\circ = \frac{156^\circ}{180^\circ} \times \pi = 2.72$$

Example:

Express the angle $\frac{\pi^c}{4}$ in degrees. Remember $\pi \text{ radian} = 180^\circ \quad \therefore 1 \text{ radian} = \frac{180^\circ}{\pi}$

$$\begin{aligned}\therefore \frac{\pi^c}{4} &= \frac{\pi}{4} \times \frac{180}{\pi} \\ &= 45^\circ\end{aligned}$$

Exercise:

Express the following angles in degrees:

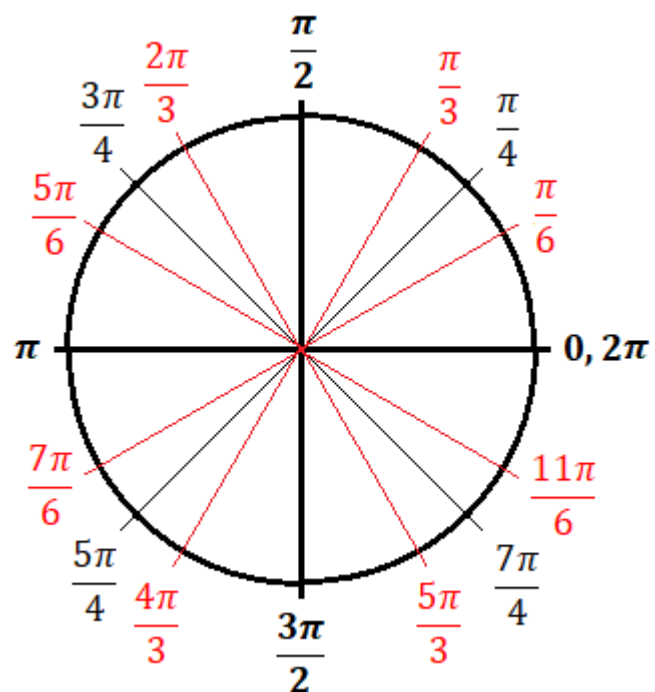
(a) $\frac{\pi}{2} = \frac{\cancel{\pi}}{2} \times \frac{180^\circ}{\cancel{\pi}} = 90^\circ$

(b) $\frac{3\pi}{2} = \frac{3\cancel{\pi}}{2} \times \frac{180^\circ}{\cancel{\pi}} = 270^\circ$

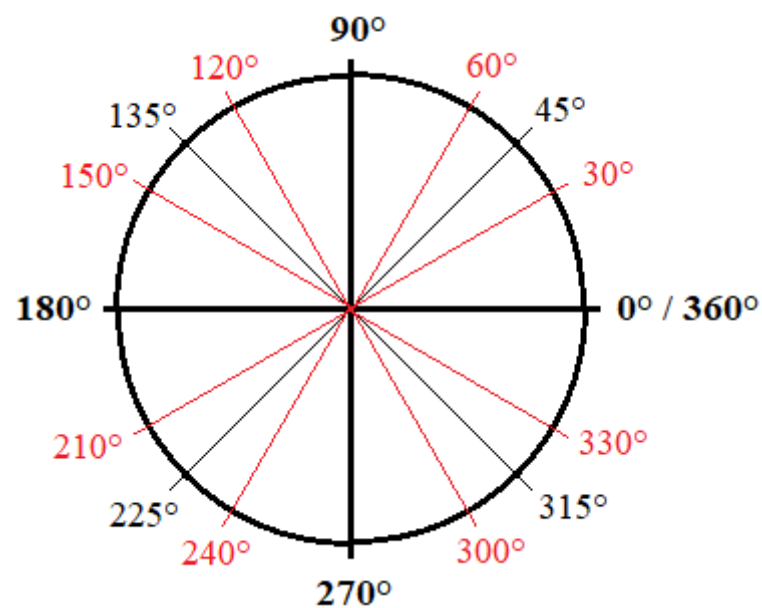
(c) $\frac{\pi}{3} = \frac{\cancel{\pi}}{3} \times \frac{180^\circ}{\cancel{\pi}} = 60^\circ$

(c) $\frac{\pi}{5} = \frac{\cancel{\pi}}{5} \times \frac{180^\circ}{\cancel{\pi}} = 36^\circ$

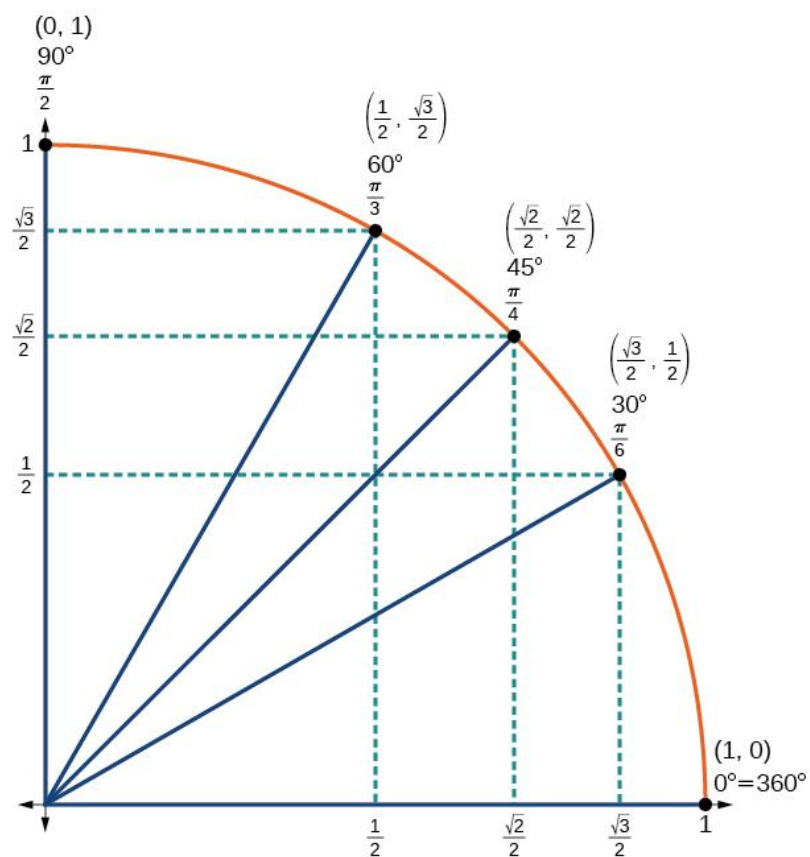
Angles in Radian



Angles in Degrees

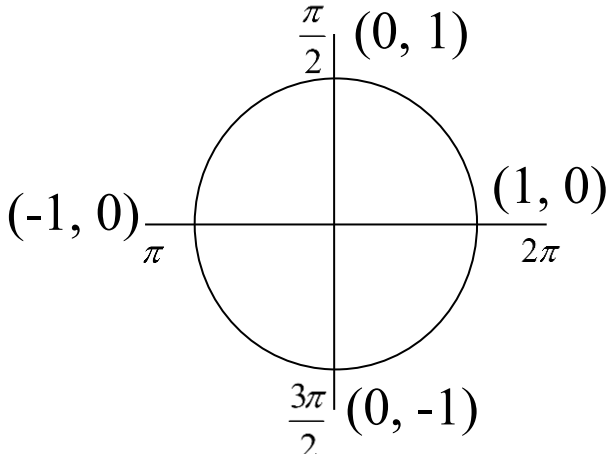


Exact Trigonometric Ratios in radians and degrees



θ Radian	θ degrees	$\sin \theta$	$\cos \theta$	$\tan \theta$
0	0°	0	1	0
$\frac{\pi}{6}$	30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
$\frac{\pi}{4}$	45°	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
$\frac{\pi}{3}$	60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2}$	90°	1	0	ND

Identifying the Quadrant of an angle

Quadrant 1	$0 < \theta < \frac{\pi}{2}$	
Quadrant 2	$\frac{\pi}{2} < \theta < \pi$	
Quadrant 3	$\pi < \theta < \frac{3\pi}{2}$	
Quadrant 4	$\frac{3\pi}{2} < \theta < 2\pi$	

Note: One revolution is 2π . Therefore, a point on the unit circle associated with an angle θ is the same as the point on the unit circle with angle $\theta \pm 2\pi$.

Exercise:

Determine the quadrant of the following angles:

- | | | | |
|-------------------------|-------------------------|--------------------------|--------------------------|
| (a) $\frac{7}{4}\pi$ Q4 | (b) 1.8π Q4 | (c) 3.25π Q3 | (d) $\frac{11\pi}{4}$ Q2 |
| (e) $\frac{2\pi}{3}$ Q2 | (f) $\frac{7}{6}\pi$ Q3 | (g) $-\frac{3\pi}{4}$ Q3 | (h) -1.9π Q1 |

Key Idea – Exact trigonometric ratio of Angles of any Magnitude (Radians)

When an angle in Q_2 , Q_3 or Q_4 has a reference angle of $\frac{\pi}{6}$, $\frac{\pi}{4}$ or $\frac{\pi}{3}$, the exact trigonometric ratios may be determined.

- Identify the quadrant in which the angle lies.
- Identify whether the trigonometric ratio is positive or negative in this quadrant.
- Determine the acute reference angle.
- Write the trigonometric ratio in terms of the reference acute angle.

Example: Find the exact value of $\sin \frac{4\pi}{3}$.

- $\frac{4\pi}{3}$ is in Q_3
- sine is negative in Q_3 .
- Reference angle = $\frac{4\pi}{3} - \pi = \frac{\pi}{3}$
- $$\begin{aligned}\sin \frac{4\pi}{3} &= -\sin \frac{\pi}{3} \\ &= -\frac{\sqrt{3}}{2}\end{aligned}$$

Exercise: Find the exact value for each of the following trigonometric ratios:

(a) $\cos\left(\frac{2\pi}{3}\right)$

- $\frac{2\pi}{3}$ is in the 2nd quadrant
- $\cos$ is negative in the 2nd quadrant

$$\bullet \theta_r = \pi - \frac{2\pi}{3} = \frac{\pi}{3}$$

$$\bullet \cos\left(\frac{2\pi}{3}\right) = -\cos\frac{\pi}{3} = -\frac{1}{2}$$

(b) $\tan\left(\frac{5\pi}{4}\right)$

- $\frac{5\pi}{4}$ is in the 3rd quadrant
- $\tan$ is positive in the 3rd quadrant

$$\bullet \theta_r = \frac{5\pi}{4} - \pi = \frac{\pi}{4}$$

$$\bullet \tan\left(\frac{5\pi}{4}\right) = \tan\left(\frac{\pi}{4}\right) = 1$$

(c) $\sin\left(\frac{11\pi}{6}\right)$

- $\frac{11\pi}{6}$ is in the 4th quadrant
- $\sin$ is negative in the 4th quadrant

$$\bullet \theta_r = 2\pi - \frac{11\pi}{6} = \frac{\pi}{6}$$

$$\bullet \sin\left(\frac{11\pi}{6}\right) = -\sin\left(\frac{\pi}{6}\right) = -\frac{1}{2}$$

Key Idea – Solving Trigonometric Equations in Radians

The steps involved in solving trigonometric equations are the same whether in degrees or radians.

Step 1: Determine the quadrants.

Step 2: Determine the reference (acute) angle.

Step 3: Determine the angles in the appropriate quadrants.

Example:

$$\text{Solve } \sin x = \frac{1}{2}, 0 \leq x \leq 2\pi$$

Step 1: Sine is positive in Q_1 and Q_2

$$\text{Step 2: Reference angle is } x_R = \frac{\pi}{6}$$

$$\text{Step 3: In } Q_1 \quad x = \frac{\pi}{6}.$$

$$\begin{aligned} \text{In } Q_2 \quad x &= \pi - \frac{\pi}{6} \\ &= \frac{5\pi}{6} \end{aligned}$$

Example:

$$\text{Solve } \cot x = -\frac{1}{\sqrt{3}}, \quad 0 \leq x \leq 2\pi.$$

$$\text{Step 1: } \cot x = -\frac{1}{\sqrt{3}}$$

$$\tan x = -\sqrt{3} \quad \tan x \text{ is negative in } Q_2 \text{ and } Q_4$$

$$\text{Step 2: Reference angle is } x_R = \tan^{-1} \sqrt{3}$$

$$= \frac{\pi}{3}$$

$$\text{Step 3: In } Q_2 \quad x = \pi - \frac{\pi}{3}$$

$$= \frac{2\pi}{3}$$

$$\text{In } Q_4 \quad x = 2\pi - \frac{\pi}{3}$$

$$= \frac{5\pi}{3}$$

$$x = \frac{2\pi}{3}, \frac{5\pi}{3}$$

Example:

Solve $3\sin x = 2\cos x$ $0 \leq x \leq 2\pi$. Give answer correct to 4 decimal places.

Step 1: $3\sin x = 2\cos x$

$$\tan x = \frac{2}{3} \quad \tan \text{ is positive in Q1 and Q3}$$

Step 2: Reference angle is $x_R = \tan^{-1}\left(\frac{2}{3}\right)$
 $= 0.5880$

Step 3: In Q1: $x = 0.5880$

In Q3: $x = \pi + 0.5880$
 $= 3.7296$

$$x = 0.5880, 3.7296$$

Note: In this example the answer cannot be given exactly in terms of π radian, therefore the answer is given in radians rounded to 4 decimal places.

Exercise:

Solve each equation for $0 \leq x \leq 2\pi$.

(a) $\cos x = \frac{1}{2}$

• $\cos$ is positive in Q1 & Q4

• $\theta_1 = \cos^{-1}\left(\frac{1}{2}\right)$
 $= \frac{\pi}{3}$

• in Q1 $\rightarrow \theta = \frac{\pi}{3}$

in Q4 $\rightarrow \theta = 2\pi - \frac{\pi}{3}$
 $= \frac{5\pi}{3}$

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

(b) $\sec x + \sqrt{2} = 0$

$$\sec x = -\sqrt{2}$$

$$\frac{1}{\cos x} = -\sqrt{2}$$

$$\cos x = -\frac{1}{\sqrt{2}}$$

• $\cos$ is negative in Q2 & Q3

• $\theta_1 = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$
 $= \frac{\pi}{4}$

• in Q2 $\rightarrow \theta = \pi - \frac{\pi}{4}$
 $= \frac{3\pi}{4}$

in Q3 $\rightarrow \theta = \pi + \frac{\pi}{4}$
 $= \frac{5\pi}{4}$

$$\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$$

Exercise:

Solve each equation for $0 \leq x \leq 2\pi$. Give answer correct to 4 decimal places.

(a) $\sin x = 0.6784$

- $\sin$ is positive in Q1 & Q2
 - $x_R = \sin^{-1}(0.6784)$
 $= 42.7187^\circ \times \frac{\pi}{180}$
 $= 0.7456^\circ$
 - in Q1 $\rightarrow \theta = 0.7456^\circ$
in Q2 $\rightarrow \theta = \pi - 0.7456^\circ$
 $= 2.3960^\circ$
- $$\theta = 0.7456^\circ, 2.3960^\circ$$

(b) $\tan^2 x - 2 \tan x = 0$

let $u = \tan x$

$$u^2 - 2u = 0$$

$$u(u - 2) = 0$$

$$\downarrow \quad \rightarrow$$
$$u = 0 \quad u = 2$$

$$\tan x = 0$$

- this is a boundary angle
- $$x = 0, \pi, 2\pi$$

$$\tan x = 2$$

- $\tan$ is positive in Q1 & Q3
- $x_R = \tan^{-1}(2)$
 $= 63.4349^\circ \times \frac{\pi}{180}$
 $= 1.1071^\circ$
- in Q1 $\rightarrow \theta = 1.1071^\circ$
in Q3 $\rightarrow \theta = \pi + 1.1071^\circ$
 $= 4.2487^\circ$

$$\theta = 0, 1.1071^\circ, \pi, 4.2487^\circ, 2\pi$$